
PROJECT PRESENTATION

StudyAI

AI-Powered Student Productivity
& Time Management Assistant

Python / Django

Node.js / Express

React + TypeScript

Docker

AI Engine

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THE CHALLENGE

Why Students Need Smarter Tools

Students juggle multiple courses, assignments, exams, and personal commitments - leading to stress, procrastination, and poor time management.

87%

of students feel
overwhelmed by workload

3.5h

average daily time
wasted on distractions

62%

procrastinate on
important deadlines

0

AI tools designed for
academic workflows

KEY INSIGHT

There is no single AI-powered tool designed specifically for the student academic workflow. StudyAI fills this gap.

OUR SOLUTION

An All-in-One AI Student Assistant

StudyAI combines intelligent task prioritization, study tracking, analytics, and proactive AI insights.

Smart Dashboard

Real-time stats, AI alerts,
Pomodoro timer & 7-day charts

Task Manager

AI-scored priorities (0-100)
with urgency-based ranking

Analytics Suite

14-day trends, focus vs.
distraction, productivity scores

AI Insights

Procrastination detection,
focus scoring & semester health

Weekly Schedule

Visual timetable with
color-coded study slots

Study Tracker

Log sessions with mood,
productivity & distractions

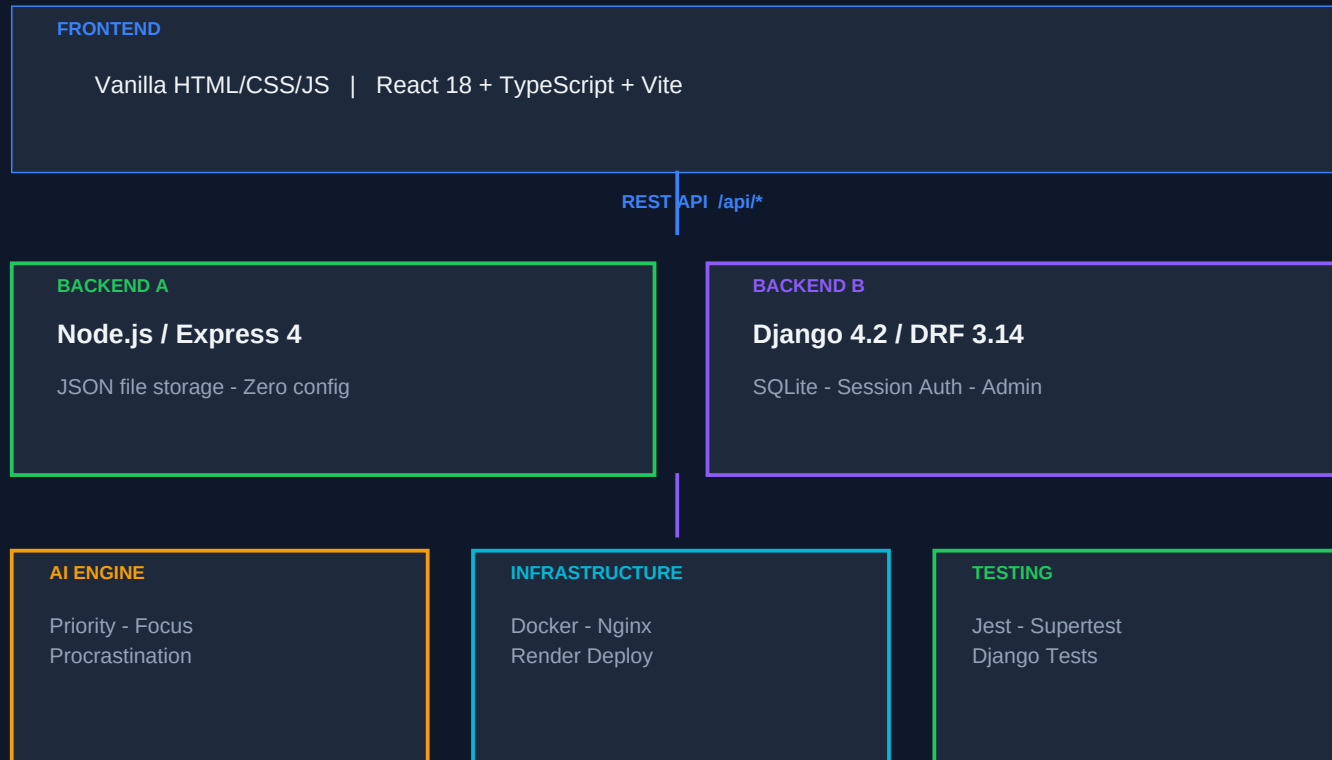
Semester Planner

Courses, exams, goals &
semester health tracking

Profile & Settings

Personal goals, Pomodoro
config & subject management

Architecture Overview



The AI Engine Behind StudyAI

Priority Scoring (0-100)

$$\text{Priority} = \text{Urgency} \times 50\% + \text{Importance} \times 35\% + \text{Effort} \times 15\%$$

Urgency = $\max(0, 100 - \text{daysLeft} \times 5)$ -> 100 if overdue

Importance = weight x 10 (weight: 1-10)

Effort = $100 - \text{estimatedHours} \times 5$

Procrastination Detection

A task is flagged when ALL THREE conditions are true:

Deadline within 48 hours

Task created > 30 minutes ago

Studied < 25% of estimated time

Focus Score

$$\text{Focus\%} = (\text{study} - \text{distraction}) / \text{study} \times 100$$

>= 80% Excellent focus!

60-79% Good - minor distractions

40-59% Moderate - try Pomodoro

< 40% High distraction risk

Semester Health Score

$$\text{Health} = \max(0, 100 - \text{overdue} \times 15 - \text{pending} \times 2)$$

Daily cron job at midnight automatically recalculates all AI metrics using node-cron (Node.js) and APScheduler (Django).

Tech Stack & Tools

FRONTEND

React 18

+ TypeScript

Vite

build tool

Vanilla JS SPA

no-build option

SVG Charts

bar / line / donut

React Router v6

lazy loading

BACKEND

Node.js 20

+ Express 4

Django 4.2

+ DRF 3.14

SQLite

+ JSON files

Session Auth

+ Django Admin

APScheduler

daily AI cron

DEVOPS & TESTING

Docker

multi-stage build

Nginx

reverse proxy

Render.com

free deployment

Jest + Supertest

unit & integration

GitHub Actions

CI/CD pipeline

Data Models & Schema

Task

title - deadline - course
type - estimatedHours
weight (1-10) - tags
completed - completedAt

StudySession

subject - duration (min)
distractionMinutes
mood - productivity (1-10)
date - notes

ScheduleSlot

title - day - startTime
endTime - subject
type (study/class/
break/exercise)

Course

name - code
instructor - credits
color

SemesterEvent

title - date - type
(exam/holiday/
submission) - course

Goal

title - targetDate
targetValue
currentValue - achieved

UserProfile

name - studyGoalHours
pomodoroWork/Break
preferredStudyTime
subjects

DailyLog

date (unique)
studyMinutes
distractionMinutes
sessions count

RESTful API Endpoints

Auth & Tasks

POST	/api/auth/register
POST	/api/auth/login
GET	/api/tasks
POST	/api/tasks
PUT	/api/tasks/:id
PATCH	/api/tasks/:id/complete
DEL	/api/tasks/:id

Study & Schedule

GET	/api/study-sessions
GET	/api/study-sessions/today
GET	/api/study-sessions/analysis
POST	/api/study-sessions
GET	/api/schedule
POST	/api/schedule
PUT	/api/schedule/:id

Analytics & Semester

GET	/api/analytics/overview
GET	/api/analytics/weekly
GET	/api/analytics/prod-trend
GET	/api/semester
POST	/api/semester/courses
POST	/api/semester/events
POST	/api/semester/goals

AI Endpoints

POST	/api/ai/analyze
GET	/api/ai/procrastination
GET	/api/ai/distraction
GET	/api/ai/time-optimize
GET	/api/ai/semester-prog
GET	/api/ai/insights
GET	/api/health

Run Anywhere in Minutes

One-Click Launch

```
# Windows
run.bat

# Linux / macOS
./run.sh
```

Auto-installs deps, seeds
demo data, starts server

Docker Production

```
docker-compose up
--build -d

curl localhost/api/health
> {"status":"ok"}
```

Multi-stage build
Non-root user, health checks

Render.com Deploy

```
git push origin main

# Auto-deploy via
# render.yaml config
```

Free tier, auto-detect config
CI/CD pipeline

Demo Data Included

6 Tasks

10 Study Sessions

7 Schedule Slots

5 Courses

5 Events

14 Days Analytics

Thank You!

StudyAI empowers students to study smarter, procrastinate less,
and achieve more - powered by intelligent AI insights.

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Core Modules

30+

API Endpoints

8

Data Models

4

AI Algorithms

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MIT Licensed

Open Source

Production Ready

Built with Python, Node.js, Django, React & Docker